

Acute GVHD

Week 6 · Ch. 43

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Four parts, from definition to salvage

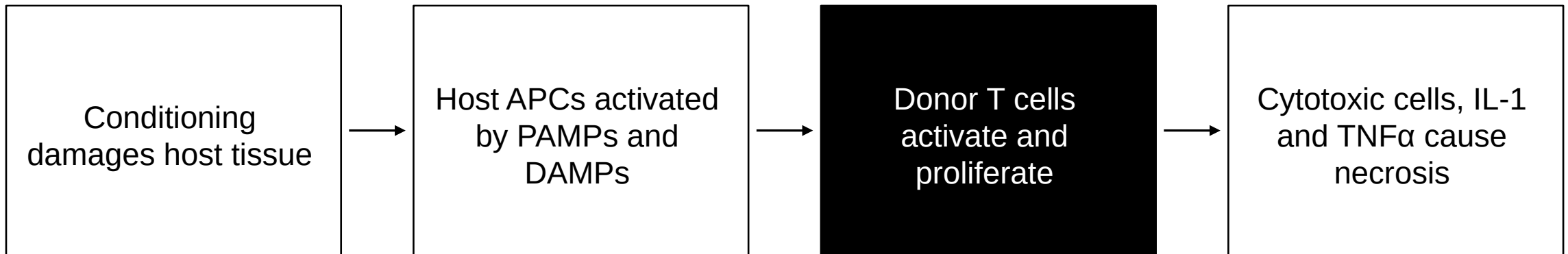
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|----------|---|---------|
| 1 | Definition, biology and risk factors
Acute vs chronic by features, three-phase cascade, who is at risk | 4 min |
| 2 | Diagnosis, staging and overall grade
Organ signs, biopsy, MAGIC organ stages, grade I–IV | 6.5 min |
| 3 | Prevention and first-line treatment
Prophylaxis trials, newer agents, methylprednisolone 2 mg/kg/day | 5.5 min |
| 4 | Steroid-refractory aGVHD and board questions
SR definition, REACH2 ruxolitinib, salvage options, 3 MCQs | 10 min |

Acute and chronic GVHD are defined by features, not day 100

Category	Timing after HCT	Acute features	Chronic features
Classic acute	< day 100	Yes	No
Persistent, recurrent or late-onset acute	> day 100	Yes	No
Classic chronic	No time limit	No	Yes
Overlap syndrome	No time limit	Yes	Yes

Late aGVHD after RIC or DLI blurs the day-100 line (Table 43.1; Filipovich 2005)

Acute GVHD follows a three-phase cascade to tissue necrosis



Brakes on the cascade: Tregs, Tr1 cells, invariant NKT cells and MDSCs. High gut microbiota diversity is protective.

HLA disparity drives acute GVHD risk

Factor	Chapter overview	Seattle, n = 2941 (Flowers 2011)
HLA mismatch (more mismatch, more severe)	Major risk	Predicts grade II–IV aGVHD
Unrelated donor	—	Predicts grade II–IV aGVHD
High-dose TBI / intense conditioning	Reported	Predicts grade II–IV aGVHD
Older patient age	Consistent risk	—
Female donor → male recipient	Consistent risk	cGVHD only, not aGVHD
Older donor; PBSC rather than BM	Reported	cGVHD only, not aGVHD
Donor alloimmunization; type of prophylaxis	Consistent risk	—
Recipient CMV seropositivity	Reported	—

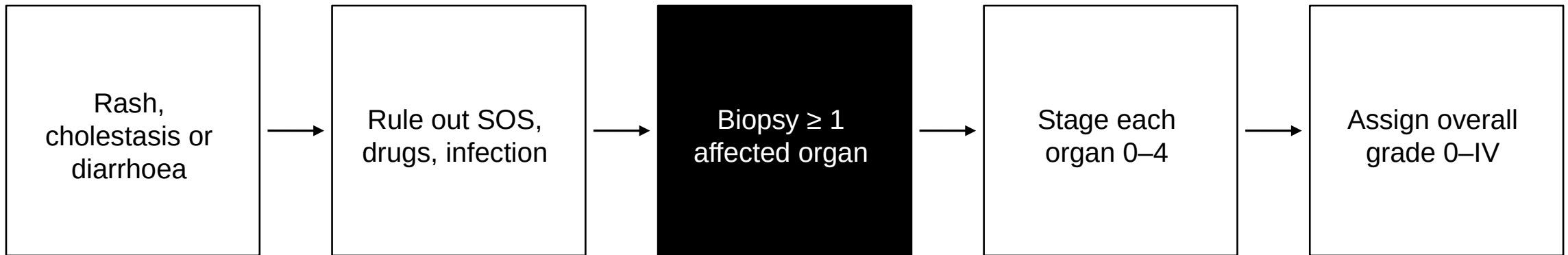
Moderate to severe aGVHD occurs in about 40% of allo-HCT recipients

Skin, liver and gut are the targets; biopsy confirms

Organ	Clinical picture	Histological hallmark
Skin	Maculopapular rash, palms and soles first; bullae if severe	Lichenoid infiltrate; basal-cell vacuolation and necrosis
Liver	Cholestasis ± jaundice; cholestatic enzymes > transaminases	Small bile duct damage; histological grade not prognostic
GI tract	Upper: anorexia, nausea, vomiting Lower: green watery diarrhoea; blood, ileus	“Exploding crypts”: necrosis of crypt epithelial cells

Shared lesion: satellite cell necrosis (immune cells beside apoptotic epithelial cells)

Diagnosis moves from organ signs to an overall grade



Speaker's summary of §43.4. The grade decides topical versus systemic therapy.

Organ stages rest on numeric cutoffs

Stage	Skin (active erythema)	Liver (bilirubin)	Upper GI	Lower GI, adult (stool/day)
0	No active rash	< 2 mg/dL	None or intermittent N/V/anorexia	< 500 mL or < 3 episodes
1	Rash < 25% BSA	2–3 mg/dL	Persistent nausea, vomiting or anorexia	500–999 mL or 3–4 episodes
2	Rash 25–50% BSA	3.1–6 mg/dL	—	1000–1500 mL or 5–7 episodes
3	Rash > 50% BSA	6.1–15 mg/dL	—	> 1500 mL or > 7 episodes
4	Erythroderma with bullae/desquamation > 5% BSA	> 15 mg/dL	—	Severe pain ± ileus, or grossly bloody stool

Table 43.4, after Harris 2016. Paediatric stool thresholds in mL/kg/day: backup.

The overall grade follows the most severe organ stage

Grade	Definition
0	No stage 1–4 in any organ
I	Skin stage 1–2, with no liver, upper GI or lower GI involvement
II	Skin stage 3 and/or liver 1 and/or upper GI 1 and/or lower GI 1
III	Liver 2–3 and/or lower GI 2–3, with skin 0–3 and/or upper GI 0–1
IV	Skin, liver or lower GI stage 4, with upper GI 0–1

Speaker's summary: upper GI alone caps at grade II; liver or lower GI stage ≥ 2 means grade $\geq$ III

Trials shaped the prophylaxis backbone

Comparison	Setting	Key result
TAC/MTX vs CSA/MTX (Ratanatharathorn 1998)	Sibling	II–IV aGVHD 32% vs 44%; no survival difference
TAC/MTX vs CSA/MTX (Nash 2000)	Unrelated	II–IV aGVHD 56% vs 74%; no survival difference
TAC/SIR vs TAC/MTX (Törlén 2016)	Phase III	Equivalent efficacy, different toxicity
ATG vs none (Finke 2009; Walker 2016)	Unrelated	Benefit mainly on chronic GVHD
ATG vs placebo (Soiffer 2017)	MUD, double-blind	Less aGVHD II–IV and cGVHD, but lower overall survival
PTCy/TAC/MMF vs TAC/MTX (Holtan 2022)	RIC, n = 431	1-year GRFS 52.7% vs 34.9% (p < 0.001)

Abatacept, vedolizumab and T-cell depletion widen prophylaxis

Abatacept

FDA-approved with a CNI and MTX for matched or 1-allele mismatched unrelated donors (Watkins 2021)

US approval cited in Ch. 43

Vedolizumab

Anti- $\alpha 4\beta 7$ integrin, selective for the gut; reduced GI aGVHD in a prospective trial (Chen 2019; Fløisand 2021)

[verify: Ch. 43 calls it phase III; cited papers are earlier-phase]

T-cell depletion

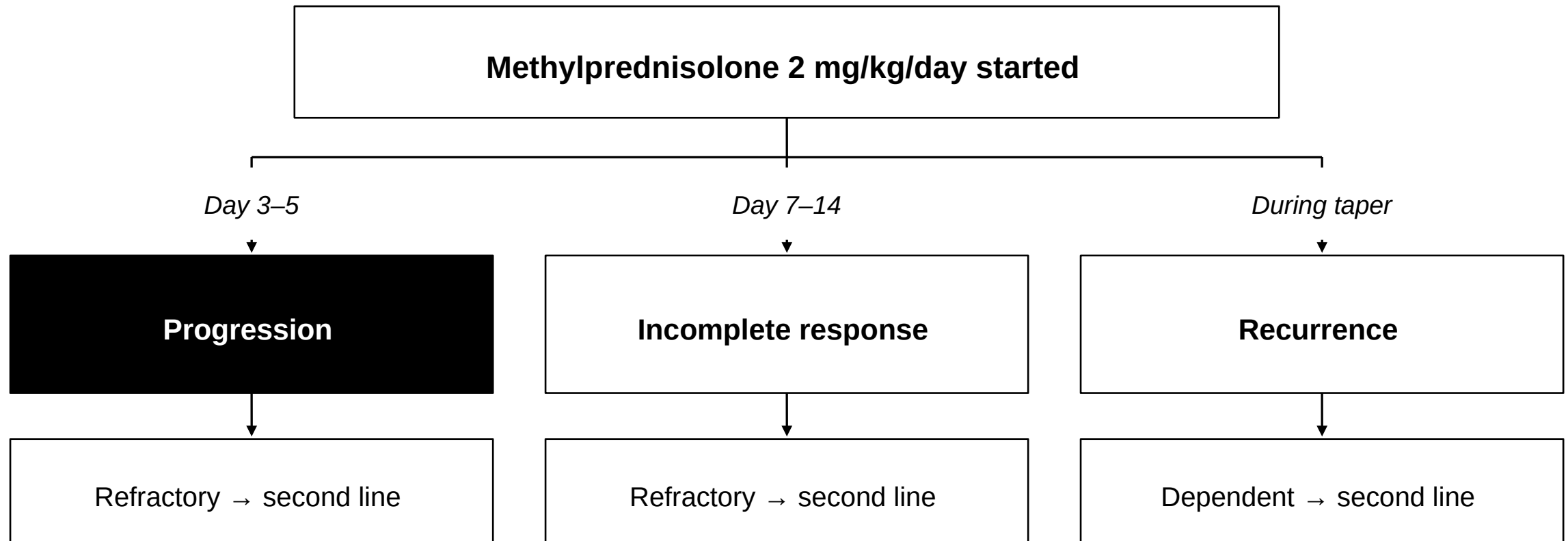
Ex vivo, CD34+ selection, ATG or MoAb, α/β depletion: strong GVHD prevention, more infection and relapse

Kept for mismatched and haploidentical grafts

Methylprednisolone 2 mg/kg/day is first-line for grade II–IV

Situation	What Ch. 43 recommends
Grade I (skin only)	Topical steroids; early systemic therapy gives no long-term gain
Mild upper GI	Lower-dose steroids plus topical therapy
Grade II–IV	Methylprednisolone 2 mg/kg/day for 7–14 days, then slow taper
Expected response	About 40–50%; lower with higher grade
Flare during taper	Raise the dose again, then taper more slowly
Add MMF or another agent?	No add-on has beaten steroids alone; more infections
Prolonged steroids	Consider anti-infective prophylaxis (viral, fungal)

Steroid failure is defined by timing and by flare on taper



Ruxolitinib is the standard for steroid-refractory acute GVHD

REACH2 (Zeiser 2020)	Ruxolitinib	Investigator's choice
Therapy	10 mg orally twice daily	ATG, ECP, MSC, MTX, MMF, mTORi, etanercept, infliximab
ORR, day 28	62%	39%
Odds ratio (95% CI)	2.64 (1.65–4.22)	
CR, day 28	34%	19%
Response, day 56	40%	22%

Phase III, n = 309, age ≥ 12, grade II–IV SR-aGVHD, 1:1. Approved by EMA and FDA.

Beyond ruxolitinib, salvage options lack consensus

Option	Evidence	Result
ECP	≥ 2 sessions/week; vs anticytokine (Jagasia 2013)	ORR 66% vs 32%; CR 54% vs 20% (Ch. 66)
MSC	Le Blanc 2008; Munneke 2016	Response > 50%; IDUNN phase III vs ruxolitinib
Pooled FMT	81 patients, GI SR-aGVHD (Malard 2023)	ORR 56%, CR 37% after median 3 doses
α1-antitrypsin	Two phase II trials	Day 28: CR 35%, ORR 60%
Refractory aGVHD	Overall outcome	About 80% mortality, worst with lower GI

Question 1: grade this patient's acute GVHD

A 45-year-old man is 30 days after a matched unrelated donor allo-HCT for AML. He has a maculopapular rash over 30% BSA, total bilirubin 4.0 mg/dL and diarrhoea of 800 mL/day. Skin and rectal biopsies confirm acute GVHD. He has no nausea. What is his overall clinical grade?

- A. Grade I
- B. Grade II
- C. Grade III
- D. Grade IV
- E. Cannot be graded until a liver biopsy is done

Answer C: liver stage 2 lifts the grade to III

C. Grade III

Skin 30% BSA = stage 2; bilirubin 4.0 mg/dL = liver stage 2 (3.1–6); 800 mL/day = lower GI stage 1 (500–999). Liver or lower GI stage 2–3 defines grade III. Grade II allows liver and lower GI stage 1 only. Staging is clinical: liver histological grading has no proven prognostic value.

Question 2: choose the next step on day 4

A 52-year-old woman develops grade III lower GI acute GVHD on day 40 after unrelated donor HCT. Methylprednisolone 2 mg/kg/day is started. On day 4 of steroids her stool output rises from 1,200 to 1,900 mL/day with fresh blood. CMV PCR and stool cultures are negative. What is the most appropriate next step?

- A. Continue the same dose and reassess on day 14
- B. Add mycophenolate mofetil to the steroids
- C. Start ruxolitinib 10 mg twice daily
- D. Increase methylprednisolone to 10 mg/kg/day
- E. Switch to topical steroids alone

Answer C: progression by day 3–5 means steroid-refractory

C. Start ruxolitinib 10 mg twice daily

Progression within 3–5 days of starting steroids defines steroid-refractory aGVHD; the 7–14-day window applies to incomplete response, not progression. In REACH2, ruxolitinib 10 mg twice daily beat investigator's choice: ORR 62% vs 39% and CR 34% vs 19% at day 28. Adding MMF to steroids failed in phase III.

Question 3: pick the correct prophylaxis statement

A 58-year-old man with AML in first complete remission will receive reduced-intensity conditioning and an 8/8 HLA-matched unrelated donor graft. The team is choosing GVHD prophylaxis. Which statement about the randomized evidence summarized in the EBMT Handbook is correct?

- A. TAC/MTX improved overall survival compared with CSA/MTX
- B. ATG in MUD HCT lowered GVHD and improved overall survival
- C. PTCy/TAC/MMF improved 1-year GRFS compared with TAC/MTX
- D. TAC/sirolimus was less effective than TAC/MTX against acute GVHD
- E. Abatacept is FDA-approved with PTCy for haploidentical donors

Answer C: PTCy improved GRFS through less acute and chronic GVHD

C. PTCy/TAC/MMF improved 1-year GRFS compared with TAC/MTX

In 431 RIC adults (Holtan 2022), PTCy/TAC/MMF gave 1-year GRFS 52.7% vs 34.9% ($p < 0.001$), from less acute and chronic GVHD. TAC/MTX lowered aGVHD without a survival gain; ATG (Soiffer 2017) lowered GVHD but OS was lower; TAC/SIR was equivalent; abatacept is approved with a CNI and MTX for unrelated donors.

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Skin and gut histology have their own grade I–IV

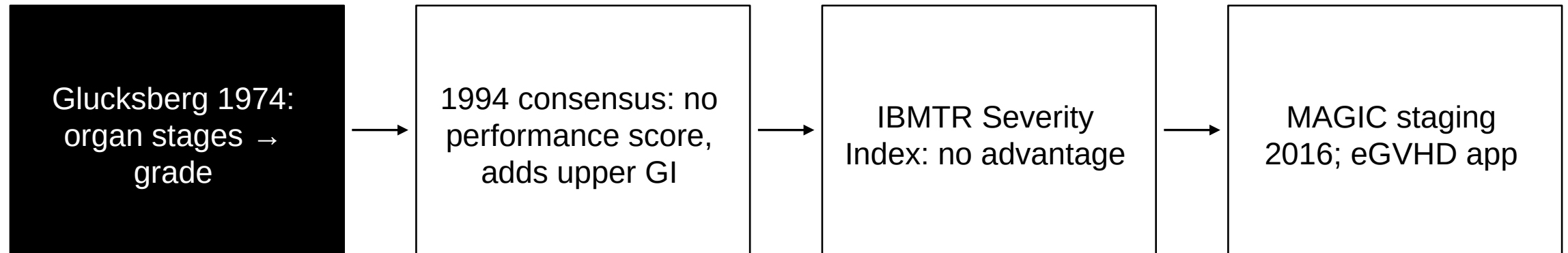
Grade	Skin	GI tract
I	Vacuolation of epidermal basal cells	Individual cell necrosis
II	Individually necrotic keratinocytes	Loss of individual crypts
III	Confluent keratinocyte necrosis forming bullae	Loss of ≥ 2 adjacent crypts with ulceration
IV	Sloughing of the epidermis	Denudation of epithelium

Liver: small bile duct damage; histological grading has no proven prognostic value

Children are staged by stool volume per kilogram

Lower GI stage	Adult (per day)	Child (per day)
0	< 500 mL or < 3 episodes	< 10 mL/kg or < 4 episodes
1	500–999 mL or 3–4 episodes	10–19.9 mL/kg or 4–6 episodes
2	1000–1500 mL or 5–7 episodes	20–30 mL/kg or 7–10 episodes
3	> 1500 mL or > 7 episodes	> 30 mL/kg or > 10 episodes
4	Severe pain ± ileus, or grossly bloody stool	Same as adult, regardless of volume

Glucksberg grading has outlasted its challengers



Glucksberg predicted survival better than the IBMTR index (Cahn 2005) and remains in regular use.

Biomarkers and tissue repair are the next frontier

Biomarkers

Validated at day 7 or at onset in multicentre cohorts; mainly flag severe GI GVHD (Vander Lugt 2013; Hartwell 2017)

Biomarker-guided trials pending

Tissue tolerance

Steroid resistance may reflect lost immunoregulation and tissue tolerance, not only resistant T cells

Hard to fix with classic IS

Gut regeneration

IL-22 and GLP-2 tested in mouse models and early clinical trials; acceptable toxicity

Needs phase III validation